

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	
Team ID	
Project Name	ServiceNow-Based Employee Laptop Request Application
Maximum Marks	4 Marks

Technical Architecture:

The architecture below shows how an employee's laptop request flows through the ServiceNow platform, from submission to approval to fulfillment.

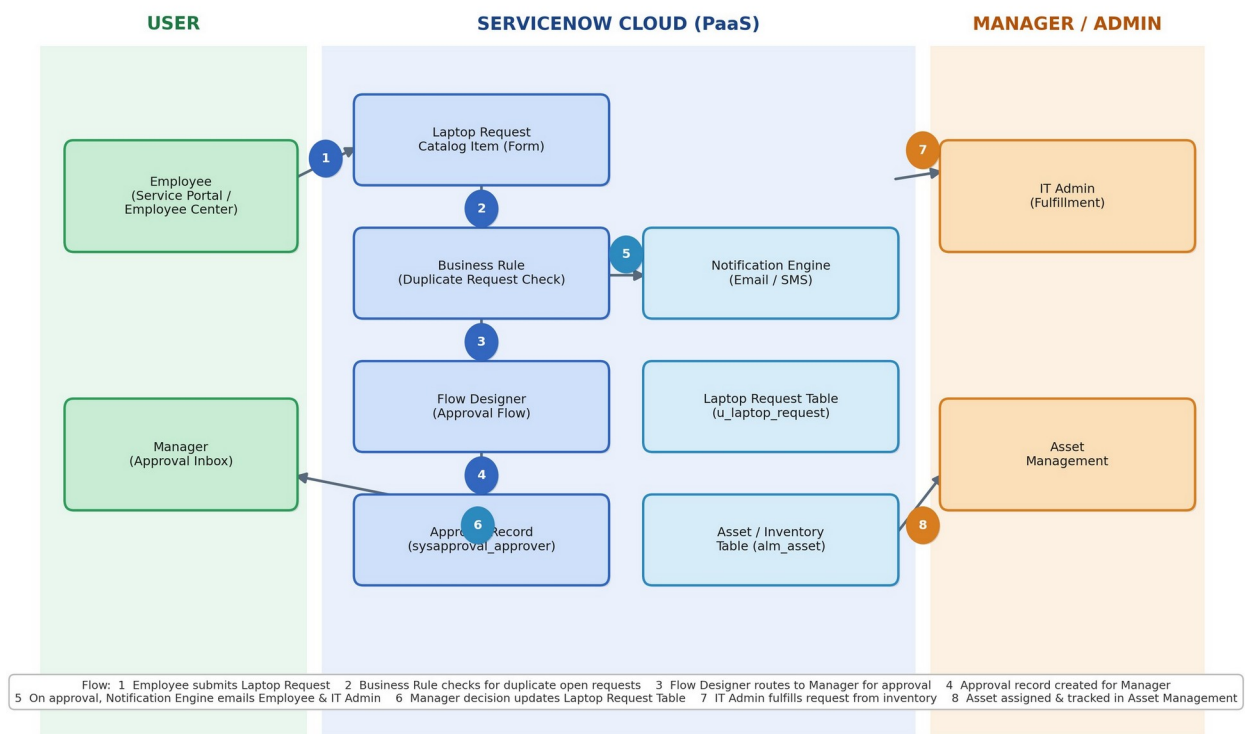


Figure 1 — Technical Architecture

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How employees submit and track laptop requests	ServiceNow Service Portal (Employee Center), HTML/CSS, AngularJS (Now UI Framework)
2.	Application Logic-1	Validates and blocks duplicate open requests before submission	ServiceNow Business Rules (server-side JavaScript)
3.	Application Logic-2	Routes the request through the approval process based on state	ServiceNow Flow Designer
4.	Application Logic-3	Generates email/SMS notifications at each state change	ServiceNow Notification Engine & Email Templates
5.	Database	Stores laptop request records, approval records, and state history	ServiceNow instance tables (u_laptop_request, sysapproval_approver)

S.No	Component	Description	Technology
6.	Cloud Database	Hosted database layer underlying the ServiceNow instance	ServiceNow multi-instance Cloud Database (PaaS)
7.	File Storage	Stores attachments such as device photos or manager justifications	ServiceNow Attachment API (sys_attachment table)
8.	External API-1	Sends outbound email/SMS notifications	SMTP Integration / Twilio Notify API
9.	External API-2	Looks up employee & manager hierarchy for routing approvals	Active Directory / LDAP Integration
10.	Machine Learning	Estimates expected approval turnaround time (Optional)	ServiceNow Predictive Intelligence
11.	Infrastructure	Application deployment (Local: Not used; Cloud: Multi-instance SaaS)	ServiceNow Cloud (PaaS) — Now Platform

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for the portal UI and styling	AngularJS (Now UI Framework component), Bootstrap
2.	Security Implementations	Access controls, encryption, and audit measures in place	Role-Based ACLs, OAuth 2.0, TLS 1.2+ encryption in transit
3.	Scalable Architecture	Justification for how the architecture scales	Multi-instance SaaS architecture with horizontal node scaling behind ServiceNow's load balancer
4.	Availability	Justification for application availability	99.8% uptime SLA backed by distributed data centers and automatic failover
5.	Performance	Design considerations for performance	Client-side caching, indexed table queries, asynchronous Flow execution

References:

<https://c4model.com/>

<https://docs.servicenow.com/bundle/washingtondc-platform-administration/>

<https://www.ibm.com/cloud/architecture>